

ON THE HOMOTOPY EQUIVALENCE BETWEEN CkNN COMPLEXES AND MANIFOLDS

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Abstract

The manifold hypothesis posits that high-dimensional real-world data often lies near a low-dimensional manifold embedded in ambient space. While this assumption underpins many techniques in machine learning and data analysis, its practical realization and validation remain challenging. In this expository work, we present a unified and accessible perspective on how the Continuous k-Nearest Neighbors (CkNN) graph construction introduced by Berry and Sauer [3] offers a mathematically consistent framework for manifold reconstruction in the context of topological data analysis (TDA). By adapting to non-uniform sampling densities and ensuring spectral convergence to the Laplace–Beltrami operator, CkNN enables the consistent recovery of topological features such as connected components and higher Betti numbers. We frame this construction as a concrete and testable instantiation of the manifold hypothesis, bridging ideas from differential geometry, spectral theory, and combinatorial topology. This note clarifies the theoretical foundations and practical significance of CkNN and, motivated by its limitations in real-world scenarios, proposes a novel variant to extend its applicability to more general settings.

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CHAPTER 1

Preliminaries and Theoretical Background

1. Introduction

The manifold hypothesis suggests that real-world data, although embedded in high-dimensional space, is concentrated near a much lower-dimensional manifold. This assumption underlies many techniques in machine learning, including dimensionality reduction, representation learning, and geometric deep learning. By positing low intrinsic dimension, it enables algorithms that are both computationally efficient and geometrically meaningful.

In topological data analysis (TDA), one aims to extract structural invariants from data—such as the number of clusters, loops, or voids—using tools from algebraic topology applied to finite point clouds. A central challenge, however, is that traditional TDA pipelines (e.g., Vietoris–Rips filtrations) depend critically on scale choices and often lack consistency guarantees: the computed homology may not converge to that of the underlying manifold as the sample size grows.

This work examines how the *Continuous k -Nearest Neighbors* (CkNN) method introduced by Berry and Sauer [3] offers a principled way to address these limitations. By constructing an unweighted graph that adapts to local sampling density, CkNN enables spectral convergence to the Laplace–Beltrami operator and thus offers a consistent path from discrete data to continuous geometry and topology.

This paper is organized in two parts. In the first, we provide an expository analysis of CkNN and its role in realizing the manifold hypothesis. In the second, we propose a modification of the CkNN framework to address specific limitations such as noise sensitivity, noncompactness, and boundary artifacts. This extension is intended to broaden the utility of CkNN while preserving its favorable convergence properties.

2. Preliminaries

We briefly recall some mathematical background relevant to our discussion. Here, we formally define a manifold as

Definition 1.1 (Manifold). *A topological space $\mathcal{M}$ is called a d -dimensional smooth manifold if every point $x \in \mathcal{M}$ has a neighborhood homeomorphic to an open subset of $\mathbb{R}^d$, and the transition maps are smooth. If $\mathcal{M} \subset \mathbb{R}^D$, we say it is embedded in $\mathbb{R}^D$.*

Additionally, a manifold is called compact if it is compact as a topological space, and connected if it is connected as a topological space. A manifold is orientable if it has a consistent choice of orientation across its charts.

For a smooth d -dimensional manifold $\mathcal{M}$ embedded in $\mathbb{R}^D$, we assume $\mathcal{M}$ is equipped with a Riemannian metric g , inducing a natural volume form dV and inner products on tangent spaces $T_p\mathcal{M}$. The Laplace–Beltrami operator $\Delta_{\mathcal{M}}$ is defined

on smooth functions $f \in C^\infty(\mathcal{M})$ as

$$\Delta_{\mathcal{M}} f = \operatorname{div}(\nabla f),$$

where ∇ and div are the gradient and divergence defined with respect to the metric g .

Let μ be a Borel probability measure supported on $\mathcal{M}$, with associated density $q \in C^2(\mathcal{M})$ with respect to the volume form dV . A point cloud $X = \{x_i\}_{i=1}^n$ is modeled as an i.i.d. sample from μ .

Definition 1.2 (Spectral Convergence). *Let $\mathcal{L}_n$ be a graph Laplacian constructed from a sample $X_n \subset \mathcal{M}$, and let $\Delta_{\mathcal{M}}$ be the Laplace–Beltrami operator on a Riemannian manifold $\mathcal{M}$. We say that $\mathcal{L}_n$ exhibits spectral convergence to $\Delta_{\mathcal{M}}$ if the eigenvalues and eigenfunctions of $\mathcal{L}_n$ converge to those of $\Delta_{\mathcal{M}}$ in the appropriate norm as $n \rightarrow \infty$.*

For analysis of geometric and spectral convergence, it is common to define a kernel $K_\varepsilon(x, y)$ and associated operators:

$$T_\varepsilon f(x) = \frac{1}{\varepsilon^{d/2}} \int_{\mathcal{M}} K\left(\frac{\|x - y\|^2}{\varepsilon}\right) f(y) q(y) dV(y),$$

as used in manifold learning literature, e.g., in diffusion maps and Laplacian Eigenmaps [1].

Given a graph $G = (V, E)$ with unweighted adjacency matrix A , the (unnormalized) graph Laplacian is $L = D - A$, where D is the degree matrix. If G is constructed from a point cloud sampled from a manifold, then L is said to converge to $\Delta_{\mathcal{M}}$ in a suitable limit if the spectrum and eigenfunctions of L converge (in an appropriate sense) to those of $\Delta_{\mathcal{M}}$.

3. The Manifold Hypothesis

Let $\mathcal{M} \subset \mathbb{R}^D$ be a compact, smooth d -dimensional Riemannian manifold. The *manifold hypothesis* posits that data sets $X = \{x_i\}_{i=1}^n \subset \mathbb{R}^D$ arising in real-world applications are sampled (possibly with noise) from a probability distribution μ supported on or near $\mathcal{M}$. More formally, we assume:

- $\mathcal{M}$ is a compact, connected, orientable Riemannian manifold without boundary;
- μ is a Borel probability measure on $\mathcal{M}$ with a smooth, strictly positive density q with respect to the Riemannian volume form;
- X is a finite i.i.d. sample from μ .

The manifold hypothesis provides a geometric prior for high-dimensional data analysis: although the ambient dimension D may be large, the intrinsic structure of the data lies in a space of much lower dimension $d \ll D$. This assumption enables both computational and statistical advantages, and underpins a variety of manifold learning techniques (e.g., Isomap, diffusion maps, Laplacian Eigenmaps) as well as topological data analysis pipelines.

The central question is: given only X , can we recover information about the geometry and topology of $\mathcal{M}$? One approach is to estimate geometric quantities such as the Laplace–Beltrami operator $\Delta_{\mathcal{M}}$, or to compute topological invariants such as the Betti numbers $\beta_k(\mathcal{M})$. However, directly testing the manifold hypothesis is statistically subtle. Fefferman, Mitter, and Narayanan [5] consider this problem in the setting of statistical hypothesis testing. They show that, under regularity

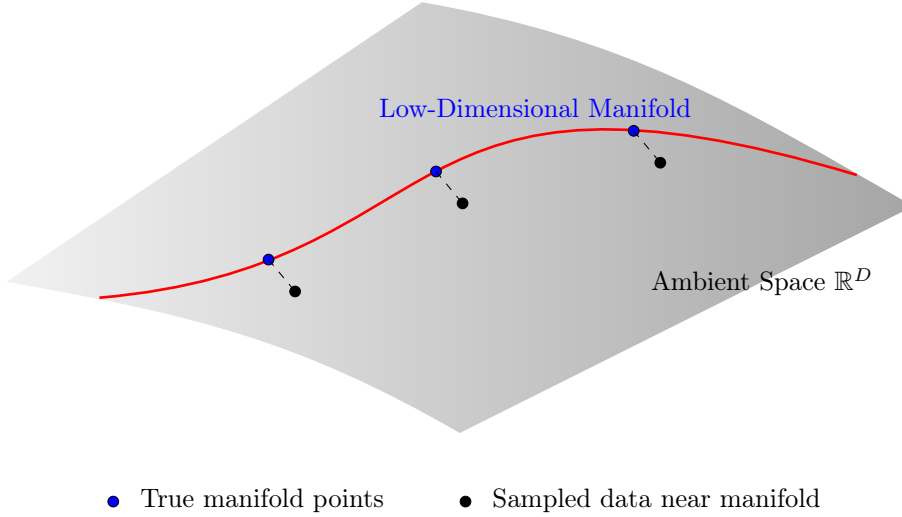


FIGURE 1. Illustration of the manifold hypothesis: high-dimensional data (black points) lie near a smooth low-dimensional manifold (red curve) embedded in ambient space.

assumptions on the reach and curvature of $\mathcal{M}$ and the noise distribution, there exist minimax optimal tests for whether a sample lies near a d -dimensional manifold. Crucially, they also establish lower bounds on the number of samples n required for reliable recovery of topological features.

In practice, consistent estimation of topological invariants from point clouds requires constructions that adapt to both the geometry of $\mathcal{M}$ and the sampling density q . As we will explore in later sections, the Continuous k -Nearest Neighbors (CkNN) graph Berry and Sauer [3] achieves this by producing a discrete Laplacian operator whose spectral properties converge to those of $\Delta_{\mathcal{M}}$, thereby linking the observed data directly to the latent manifold structure.

4. Topological Data Analysis and Persistent Homology

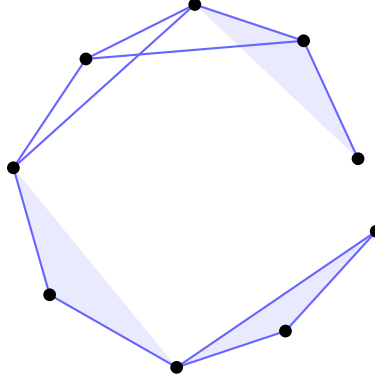
Topological Data Analysis (TDA) provides a framework for extracting structural information about a data set by leveraging techniques from algebraic topology. Given a point cloud $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^D$, one seeks to understand its underlying topological features: connected components, loops, voids, and more generally, k -dimensional holes. A common approach is to construct a family of combinatorial complexes parameterized by a scale parameter and compute the homology groups of each.

Let $\mathcal{K}_\varepsilon$ be a simplicial complex constructed from X at scale $\varepsilon > 0$ – typically via the Vietoris–Rips (VR) complex:

$$\text{VR}(X, \varepsilon) := \{\sigma \subset X : \|x_i - x_j\| \leq \varepsilon \ \forall \ x_i, x_j \in \sigma\}.$$

This yields a nested filtration:

$$\text{VR}(X, \varepsilon_1) \subseteq \text{VR}(X, \varepsilon_2) \subseteq \dots \quad (\varepsilon_1 < \varepsilon_2 < \dots).$$



Vietoris-Rips complex with 0-, 1-, and 2-simplices

FIGURE 2. Sample Vietoris-Rips complex built over a noisy circular point cloud. Triangles represent 2-simplices formed when triplets of nearby points are mutually connected. The loop-like structure is preserved topologically.

The k -th persistent homology is then computed via the induced maps on homology:

$$H_k(\text{VR}(X, \varepsilon_1)) \rightarrow H_k(\text{VR}(X, \varepsilon_2)) \rightarrow \cdots.$$

Each homology class $\alpha \in H_k(\text{VR}(X, \varepsilon_i))$ has a birth time ε_i and may die at ε_j if it maps to zero in $H_k(\text{VR}(X, \varepsilon_j))$ but not before.

These intervals $[\varepsilon_{\text{birth}}, \varepsilon_{\text{death}})$ form the k -dimensional persistence diagram.

However, while persistent homology provides a powerful multi-scale summary of topological features, it suffers from key theoretical limitations. In particular, the VR complex is sensitive to the choice of scale ε , and there is no canonical scale at which one recovers the true topology of the underlying manifold. More importantly, as noted by Berry and Sauer [3], the VR construction lacks a consistent limit: the persistent homology does not converge (in any strong sense) to the homology of the underlying manifold $\mathcal{M}$ as $n \rightarrow \infty$, unless certain weighted or density-adaptive methods are introduced.

Let us denote by $\mathcal{M}$ a compact d -dimensional Riemannian manifold embedded in $\mathbb{R}^D$, and suppose $X_n = \{x_1, \dots, x_n\}$ is an i.i.d. sample from a distribution supported on or near $\mathcal{M}$. Then the fundamental question is whether

$$\lim_{n \rightarrow \infty} H_k(\text{VR}(X_n, \varepsilon_n)) \cong H_k(\mathcal{M}),$$

for some suitable sequence ε_n . In general, this fails: no single scale ε can consistently recover all Betti numbers of $\mathcal{M}$ without either under- or over-connecting components depending on local sampling density. This phenomenon is exacerbated in nonuniformly sampled data sets, where fixed- ε constructions misrepresent both local and global topology (cf. Fig. 1 of [3]).

These issues highlight the need for consistent, density-adaptive constructions that preserve both geometric and topological information. The Continuous k -Nearest Neighbors (CkNN) graph, introduced by Berry and Sauer [3], was designed precisely to resolve this inconsistency. It defines an unweighted graph that reflects a consistent geometry, with the remarkable property that the unnormalized graph

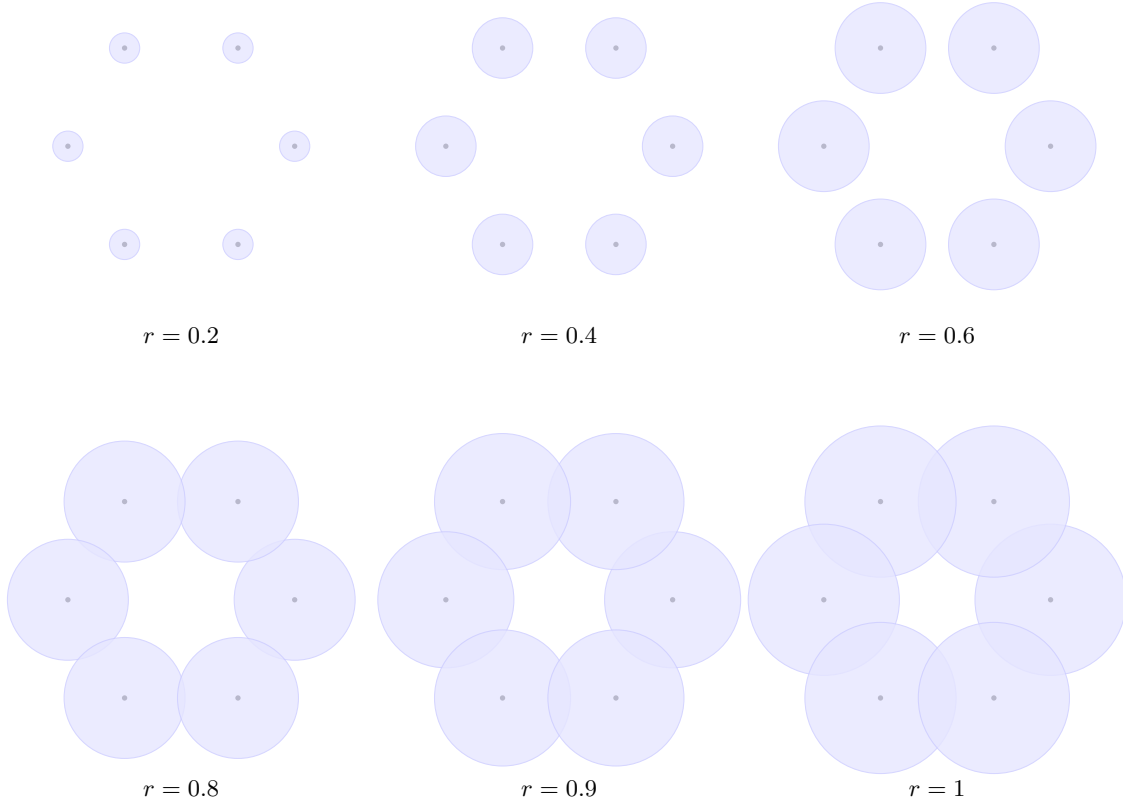


FIGURE 3. Rips filtration: as the radius increases, higher-dimensional simplices appear and topological features emerge and vanish.

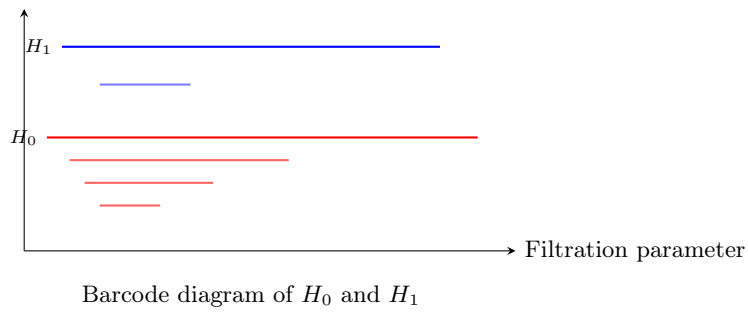


FIGURE 4. Persistence barcode showing the birth and death of topological features across the filtration. Long bars represent persistent, meaningful structures (e.g., a loop in H_1); short bars correspond to topological noise.

Laplacian converges spectrally to the Laplace–Beltrami operator on $\mathcal{M}$. This spectral convergence is the gateway to topological consistency, connecting the discrete Laplacian to the continuous Laplace–de Rham complex and ultimately to simplicial homology via spectral exterior calculus [2].

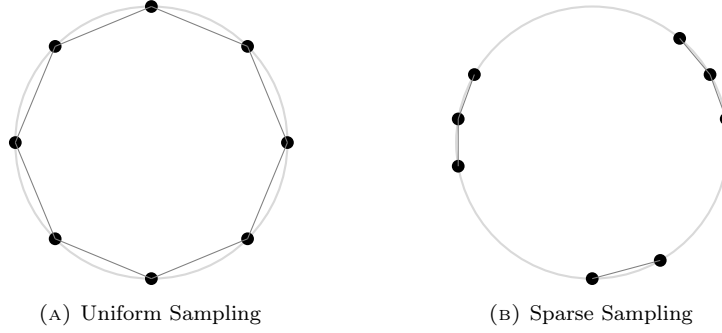


FIGURE 5. Illustration of fixed-distance (ε) graphs applied to a noisy sampling of the circle.

Figure 5a is a graphical example of a uniform sampling preserves connectivity, which successfully capturing the loop structure.

On the other hand, figure 5b is a graphical example of a sparse or uneven sampling leads to under-connection, breaking the loop and misrepresenting the underlying topology. This motivates density-aware constructions like CkNN.

Thus, while traditional VR-based persistent homology captures topological features at multiple scales, it lacks a unifying, convergent structure. CkNN aims to provide this structure: a single graph construction that encodes the correct topology at once, free from the scale sensitivity inherent to conventional TDA pipelines.

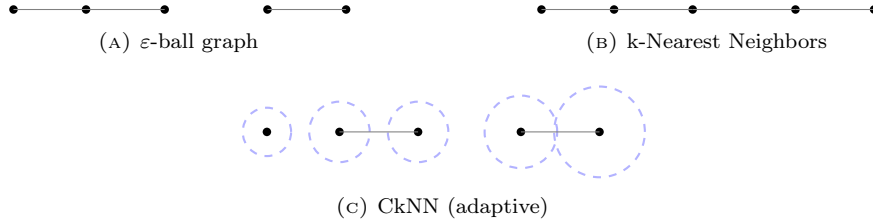


FIGURE 6. Comparison of graph constructions on nonuniformly spaced data points. **Left:** ε -ball graph fails to connect across sparse regions. **Center:** kNN connects all points, but ignores density. **Right:** CkNN adapts edge thresholds to local density using a scaling based on $r_k(x)$.

5. The CkNN Framework

Let $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^D$ be a finite sample drawn i.i.d. from a Borel probability measure μ supported on a compact d -dimensional Riemannian manifold $\mathcal{M} \subset \mathbb{R}^D$, with smooth density q with respect to the volume form dV . The goal is to construct a graph G_n on X such that the unnormalized graph Laplacian L_n converges, in a spectral sense, to the Laplace–Beltrami operator $\Delta_{\mathcal{M}}$ on $\mathcal{M}$.

The Continuous k -Nearest Neighbors (CkNN) method, introduced in Berry and Sauer [3], defines an unweighted graph whose edge set is determined by both pairwise distances and local sampling density. For each point $x_i \in X$, let $r_k(x_i)$ denote the Euclidean distance to its k -th nearest neighbor.

Definition 1.3 (CkNN Graph). Let $X_n = \{x_1, \dots, x_n\} \subset \mathbb{R}^D$ be a set of sample points. Fix $k \in \mathbb{N}$, and let $r_k(x_i)$ denote the distance from x_i to its k -th nearest neighbor. Given $\delta > 0$, the Continuous k -Nearest Neighbors (CkNN) graph $G_n = (X_n, E_n)$ is constructed by connecting points x_i and x_j if

$$\|x_i - x_j\| < \delta \sqrt{r_k(x_i) r_k(x_j)},$$

where E_n is the edge set of G_n . The unweighted adjacency matrix A_n of G_n has entries

$$A_{ij} = \begin{cases} 1 & \text{if } \|x_i - x_j\| < \delta \sqrt{r_k(x_i) r_k(x_j)} \\ 0 & \text{otherwise} \end{cases}.$$

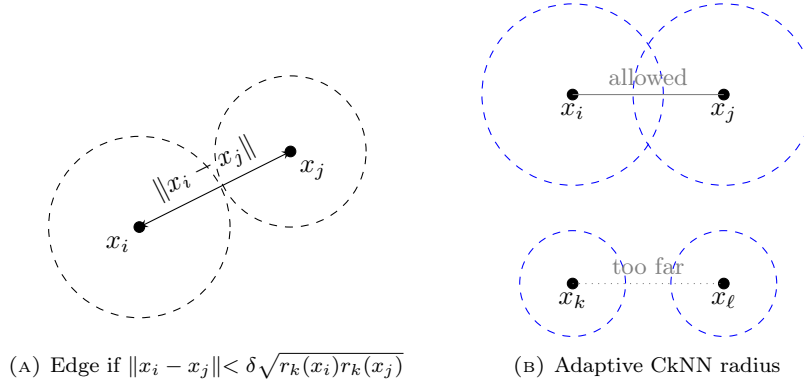


FIGURE 7

Figure 7a represents the CkNN graph construction criterion. An edge is placed between x_i and x_j if their distance is less than $\delta \sqrt{r_k(x_i) r_k(x_j)}$, where $r_k(x_i)$ is the distance from x_i to its k -th nearest neighbor. This criterion adapts to local sampling density, enabling consistent topological recovery even under nonuniform sampling.

Figure 7b is an illustration of CkNN's adaptive edge rule. In high-density regions (left), threshold radii are larger, permitting edges. In sparse regions (right), edge thresholds are tighter, restricting connectivity.

The adaptive construction has two key features:

- It adapts to local sampling density via $r_k(x)$, effectively normalizing distances to avoid over- or under-connecting in high- or low-density regions.
- It produces an unweighted adjacency matrix A_n whose associated Laplacian $L_n = D_n - A_n$ (where D_n is the diagonal degree matrix) retains topological structure in the limit $n \rightarrow \infty$.

Berry and Sauer [3] prove that, under mild regularity conditions, L_n converges spectrally to $\Delta_{\mathcal{M}}$ up to a multiplicative constant. More precisely:

THEOREM 1.4 (Spectral Convergence of CkNN [2]). Let $f_1, \dots, f_m$ be the first m eigenfunctions of the unnormalized graph Laplacian L_n on the CkNN graph G_n , with corresponding eigenvalues $\lambda_1^{(n)} \leq \dots \leq \lambda_m^{(n)}$. Then, with high probability as $n \rightarrow \infty$,

$$\lambda_j^{(n)} \rightarrow C \cdot \lambda_j \quad \text{and} \quad f_j^{(n)} \rightarrow f_j,$$

for $j = 1, \dots, m$, where λ_j and f_j are the eigenvalues and eigenfunctions of the Laplace–Beltrami operator $\Delta_{\mathcal{M}}$, and $C > 0$ is a constant depending on δ , k , and the density q .

In particular, this means that CkNN recovers not only the spectrum of $\Delta_{\mathcal{M}}$ (up to scaling), but also its eigenfunctions — which encode the intrinsic geometry of the manifold. Since the Laplace–Beltrami operator plays a central role in spectral geometry and in the Hodge theory underlying topological invariants, this result opens a rigorous path from point-cloud data to manifold structure.

Furthermore, because the CkNN graph is unweighted, it can be used to build simplicial complexes (e.g., clique complexes) for TDA. This enables one to compute Betti numbers using persistent homology, while being guaranteed (in principle) that the resulting homological features are consistent in the large-sample limit.

Compared to ε -ball graphs and traditional k NN graphs, CkNN enjoys consistency under nonuniform sampling. Fixed- ε graphs fail to adapt to changing point density and lead to topological distortions, while standard k NN graphs are typically weighted and thus incompatible with standard persistent homology software. CkNN provides a principled solution to both problems.

Finally, the parameter δ provides a natural scale control. For appropriate choices of δ , the CkNN graph captures the topological regime of the manifold, and Berry and Sauer [3] propose heuristic procedures for tuning δ in practice (see Berry and Sauer [3], §6).

Thus, CkNN provides a consistent and geometrically faithful way to bridge discrete data and continuous manifold structure. In the next section, we interpret this as a partial realization of the manifold hypothesis.

6. Manifold Reconstruction and the Hypothesis

Let $\mathcal{M} \subset \mathbb{R}^D$ be a compact, connected, smooth d -dimensional Riemannian manifold without boundary, and let μ be a Borel probability measure on $\mathcal{M}$ with smooth, strictly positive density q with respect to the Riemannian volume form dV . The manifold hypothesis posits that a data set $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^D$ is sampled i.i.d. from μ , and that $\mathcal{M}$ encodes the intrinsic geometric and topological structure of the data.

The central task of manifold reconstruction is to infer geometric and topological invariants of $\mathcal{M}$ from the finite sample X . Specifically, one seeks to recover:

- The geometry of $\mathcal{M}$, as described by the Riemannian metric g , and more tractably, the Laplace–Beltrami operator $\Delta_{\mathcal{M}}$;
- The topology of $\mathcal{M}$, as given by homological invariants such as the Betti numbers $\beta_k(\mathcal{M})$;
- A discrete structure (e.g., a graph or simplicial complex) on X that approximates $\mathcal{M}$ in a consistent way.

We interpret the CkNN method as providing a statistically consistent estimator of both the differential and topological structure of $\mathcal{M}$, thereby offering a constructive realization of the manifold hypothesis.

6.1. From Graph Laplacians to $\Delta_{\mathcal{M}}$. Let $G_n = (X, E_n)$ be the CkNN graph with adjacency matrix A_n and unnormalized graph Laplacian $L_n = D_n - A_n$. Define

the function space $\mathcal{F}_n = \{f : X \rightarrow \mathbb{R}\}$, and equip it with the discrete inner product

$$\langle f, g \rangle_n = \frac{1}{n} \sum_{i=1}^n f(x_i)g(x_i).$$

The Laplacian L_n defines a symmetric operator on $\mathcal{F}_n$, and its spectrum $\{\lambda_j^{(n)}\}$ and eigenvectors $\{f_j^{(n)}\}$ are discrete analogues of the eigenvalues and eigenfunctions of $\Delta_{\mathcal{M}}$.

Berry and Sauer [3] show that, under mild assumptions, L_n converges spectrally to $C \cdot \Delta_{\mathcal{M}}$ as $n \rightarrow \infty$, for some constant $C > 0$. That is, for each fixed $j \in \mathbb{N}$,

$$\lambda_j^{(n)} \rightarrow C \cdot \lambda_j \quad \text{and} \quad f_j^{(n)} \rightarrow f_j \in L^2(\mathcal{M}, q \, dV),$$

where λ_j and f_j are the j -th eigenvalue and eigenfunction of $\Delta_{\mathcal{M}}$ acting on $L^2(\mathcal{M}, q \, dV)$. This is convergence in the sense of operator spectra (specifically, norm-resolvent or strong resolvent convergence), and it implies that L_n preserves key geometric information about $\mathcal{M}$.

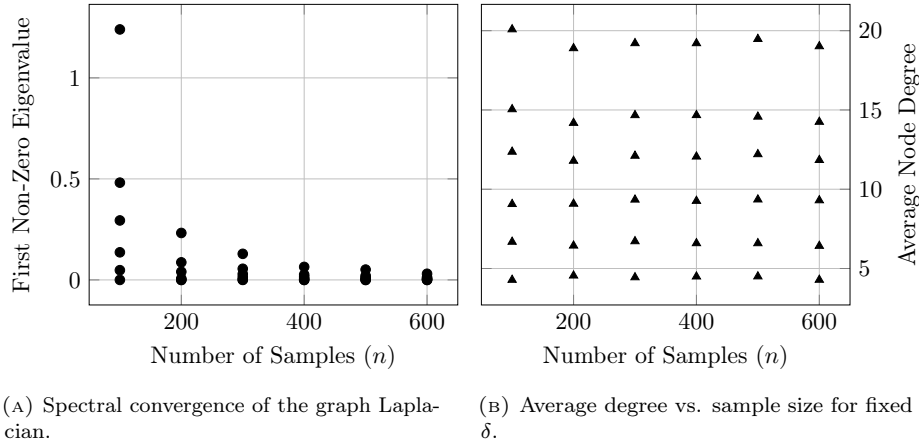


FIGURE 8. Empirical behavior of CkNN graphs under increasing sample size.

Figure 8a is a graph of the first nonzero Laplacian eigenvalue of the CkNN graph vs. n , showing convergence toward the Laplace–Beltrami spectrum.

Figure 8b is a graph of the average node degree as a function of n , holding δ fixed.

These plots illustrate the combinatorial and spectral stability of CkNN constructions under refinement, consistent with theoretical expectations.

6.2. Spectral Geometry and Topology. The Laplace–Beltrami operator $\Delta_{\mathcal{M}}$ encodes more than just geometry. Via Hodge theory, its eigenstructure governs the de Rham cohomology of $\mathcal{M}$. In particular:

- The dimension of the kernel of the Hodge Laplacian on k -forms, $\ker \Delta_k$, is isomorphic to the k -th Betti number $\beta_k(\mathcal{M})$.
- For $k = 0$, $\ker \Delta_0$ corresponds to the space of constant functions, and its dimension equals the number of connected components of $\mathcal{M}$.

The unnormalized graph Laplacian L_n acts on functions (0-forms) and provides an estimator for Δ_0 . More generally, using tools from spectral exterior calculus (SEC), one can define discrete Hodge Laplacians $\Delta_k^{(n)}$ on combinatorial k -forms, and under suitable constructions (e.g., clique complexes from CkNN graphs), these discrete operators converge to their continuous counterparts Δ_k .

Thus, via spectral convergence, CkNN enables consistent estimation of the Betti numbers $\beta_k(\mathcal{M})$ from finite samples:

$$\beta_k(\mathcal{M}) = \dim \ker \Delta_k = \lim_{n \rightarrow \infty} \dim \ker \Delta_k^{(n)}.$$

This bridges geometric learning and topological data analysis through a shared spectral framework.

6.3. Consistency and Realization of the Hypothesis. We can now make precise what it means for CkNN to realize the manifold hypothesis.

Let $X_n = \{x_1, \dots, x_n\}$ be an i.i.d. sample from μ , and let L_n be the unnormalized graph Laplacian of the CkNN graph G_n . Suppose we define $H_k^{(n)}$ to be the k -th persistent homology of a simplicial complex (e.g., the clique complex) built from G_n .

Under the assumptions of Berry and Sauer [3], we have the convergence:

$$\lim_{n \rightarrow \infty} H_k^{(n)} \cong H_k(\mathcal{M}),$$

provided that the parameter δ is chosen in the topological regime (i.e., δ lies in an interval where the CkNN graph captures the correct topological structure). This defines topological consistency, a property rarely achieved by fixed-radius or standard k NN graphs.

Moreover, since CkNN yields a single graph that preserves all homology generators simultaneously (unlike persistence which tracks generators across scales), it offers a strong practical interpretation of the manifold hypothesis: the existence of a finite sample X_n from which one can recover all topological features of $\mathcal{M}$ via a consistent discrete structure.

Thus, the CkNN method provides a partial realization of the manifold hypothesis in the following sense:

- Given: i.i.d. samples from a smooth measure on a compact manifold,
- Construct: a CkNN graph with appropriate parameters,
- Recover: the correct topological invariants and the spectrum of $\Delta_{\mathcal{M}}$,
- Conclude: consistency with the manifold hypothesis.

This not only affirms the relevance of the hypothesis in practical data analysis but also offers a concrete pipeline for its operationalization in topological and spectral learning.

While CkNN achieves theoretical consistency under compactness, smoothness, and uniform sampling conditions, real-world data sets often exhibit properties that violate these assumptions—such as noise, boundaries, sparsity, or variable curvature. These challenges motivate potential modifications of the CkNN framework that preserve its favorable convergence guarantees while extending its applicability. In the next section, we introduce a novel extension of CkNN designed to address these limitations.

7. Broader Context and Related Work

The reconstruction of geometric and topological structure from point cloud data has been a central pursuit in manifold learning, spectral geometry, and topological data analysis (TDA). The Continuous k -Nearest Neighbors (CkNN) method sits at the intersection of these fields, resolving critical limitations of earlier approaches through a blend of geometric adaptivity and spectral consistency.

This section situates CkNN within the broader landscape of data-driven manifold reconstruction and TDA, emphasizing both the theoretical advances and practical advantages over prior methods.

7.1. Spectral Methods in Manifold Learning. Early manifold learning methods, such as Laplacian Eigenmaps [1] and Diffusion Maps [4], approximate the geometry of a manifold $\mathcal{M} \subset \mathbb{R}^D$ by constructing a weighted graph G_n on a point cloud $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^D$, typically via a fixed-bandwidth kernel:

$$K_\varepsilon(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{\varepsilon}\right),$$

with edge weights $w_{ij} = K_\varepsilon(x_i, x_j)$, and graph Laplacian

$$L_n = D_n - W_n,$$

where $W_n = (w_{ij})$ and D_n is the diagonal degree matrix. These constructions yield convergence:

$$L_n \rightarrow \Delta_{\mathcal{M}} \quad (\text{up to normalization}),$$

under strong regularity assumptions on the sampling distribution μ and manifold $\mathcal{M}$. However, these graphs are weighted, and thus not directly compatible with persistent homology, which typically assumes unweighted complexes (e.g., Vietoris–Rips or clique complexes built from adjacency graphs).

Moreover, the bandwidth parameter ε must shrink with n at a precise rate to ensure consistency (cf. Belkin and Niyogi, 2007). The choice of ε affects both spectral accuracy and graph connectivity, leading to a tension between geometric fidelity and topological interpretability.

In contrast, the CkNN method bypasses these limitations:

- It constructs unweighted graphs using a density-adaptive local scale derived from k -NN distances.
- It achieves spectral convergence to $\Delta_{\mathcal{M}}$ with fixed k and fixed δ , under mild assumptions.
- It enables the construction of simplicial complexes compatible with persistent homology, while maintaining geometric consistency.

Thus, CkNN unifies the spectral convergence guarantees of Laplacian-based methods with the combinatorial tractability required for TDA.

7.2. Limitations of Traditional TDA Pipelines. Topological Data Analysis has traditionally relied on scale-parameterized constructions such as the Vietoris–Rips complex $\text{VR}(X, \varepsilon)$, Čech complex, or witness complexes. These complexes yield a filtration:

$$\mathcal{K}_{\varepsilon_1} \subseteq \mathcal{K}_{\varepsilon_2} \subseteq \dots,$$

and homology is computed over each $\mathcal{K}_{\varepsilon_i}$, producing persistent homology and corresponding persistence diagrams or barcodes.

The appeal of persistent homology lies in its robustness to noise and its multiscale character. However, the following theoretical limitations apply:

- (1) Lack of a consistent limit: As shown by Berry and Sauer [3], persistent homology computed from $\text{VR}(X_n, \varepsilon_n)$ does not, in general, converge to $H_k(\mathcal{M})$ as $n \rightarrow \infty$.
- (2) Scale sensitivity: No single ε recovers all Betti numbers reliably, especially under non-uniform sampling.
- (3) Incompatibility with spectral convergence: The VR complex lacks a spectral operator whose eigenstructure approximates $\Delta_{\mathcal{M}}$.

While some extensions, such as density-sensitive filtrations or weighted complexes, attempt to address these issues, they introduce significant algorithmic and theoretical complexity, and often rely on heuristics or lack convergence guarantees.

CkNN resolves these challenges by producing a single unweighted graph whose spectrum converges to that of $\Delta_{\mathcal{M}}$, thereby enabling a topologically consistent complex and a geometrically faithful Laplacian simultaneously.

7.3. Statistical Testing and the Manifold Hypothesis. Fefferman, Mitter, and Narayanan [5] formulate the manifold hypothesis as a statistical testing problem: given i.i.d. samples from a distribution P on a Hilbert space H , determine whether there exists a d -dimensional submanifold $\mathcal{M}$ with bounded volume and positive reach such that the mean squared distance

$$L(\mathcal{M}, P) = \int_H \text{dist}(x, \mathcal{M})^2 dP(x),$$

is small.

They prove minimax lower bounds for this problem, showing that the number of samples required for consistent hypothesis testing grows at least as

$$n \gtrsim \left(\frac{1}{\tau^d \varepsilon^{d/2}} + \frac{1}{\tau^d} \right),$$

where τ is the reach of the manifold and ε is the desired approximation error. These results imply that manifold learning is fundamentally sample-complex, especially in high intrinsic dimension or under small reach.

While their approach yields provable guarantees on manifold detection, it does not offer an explicit reconstruction method. In contrast, CkNN provides an algorithmic construction of the manifold's topology and geometry from finite samples, sidestepping the need for hypothesis testing in favor of constructive recovery.

Thus, CkNN complements the Fefferman–Mitter–Narayanan framework: the latter offers impossibility results and sample complexity bounds; the former supplies an implementable estimator consistent with the same underlying assumptions.

7.4. Spectral Exterior Calculus and Higher Homology. Recent work in spectral exterior calculus (SEC) [2] aims to extend Laplacian-based techniques to higher-degree forms, thereby enabling approximation of the Hodge Laplacians Δ_k acting on differential k -forms. These operators govern the de Rham cohomology of $\mathcal{M}$, and their kernel dimensions yield the higher Betti numbers:

$$\beta_k(\mathcal{M}) = \dim \ker \Delta_k.$$

While traditional TDA computes homology through combinatorial boundary maps, SEC provides an analytic pathway using function spaces and spectral operators.

CkNN provides the discretization foundation for SEC by supplying consistent approximations of Δ_0 and (conjecturally) Δ_k via discrete exterior operators constructed from the unweighted CkNN graph.

Hence, CkNN can be viewed as the discrete substrate upon which analytic methods like SEC are built, further bridging the gap between combinatorial and spectral topology.

8. Examples

Example 1.5 (CkNN on the Circle S^1). *Let $\mathcal{M} = S^1 \subset \mathbb{R}^2$ be the unit circle. Consider a sample of n points drawn uniformly in angle from S^1 . Construct the CkNN graph using a fixed k and suitable δ . For sufficiently large n , the resulting graph forms a near-cyclic structure. Its clique complex approximates a triangulated circle, recovering both the homology and (empirically) the homotopy type of S^1 .*

Example 1.6 (Failure of ε -Ball Graphs under Non-Uniform Sampling). *Consider a non-uniform sample from a one-dimensional manifold (e.g., a circle with dense points on one arc and sparse points on another). An ε -ball graph with fixed radius may fail to connect sparse regions or may overconnect dense ones, distorting the topology. CkNN adaptively rescales local distances, allowing it to preserve connectivity and loop structure more robustly.*

In the next chapter, we build upon this foundation to prove that, under mild conditions, the CkNN complex is homotopy equivalent to the underlying manifold.

CHAPTER 2

Homotopy Equivalence of CkNN Complexes

1. Motivation and Overview

Let $\mathcal{M} \subset \mathbb{R}^D$ be a compact, smooth Riemannian manifold without boundary, and let $X_n = \{x_1, \dots, x_n\}$ be an i.i.d. sample from the uniform distribution on $\mathcal{M}$. Construct the unweighted CkNN graph $G_n = (X_n, E_n)$ with parameter $\delta > 0$, and define K_n to be the clique complex (or a related Vietoris–Rips-style complex) built from G_n .

While prior work has established convergence of the spectrum of the CkNN graph Laplacian and asymptotic recovery of homological features, our goal here is to show that, under suitable conditions, the full homotopy type of $\mathcal{M}$ is captured with high probability by the simplicial complex K_n .

2. Formal Setup and Problem Statement

In the preceding sections, we showed how the CkNN graph provides a consistent estimator for both the geometry and topology of a manifold from sampled data, as formalized through spectral convergence and persistent homology. However, while CkNN achieves convergence of homological invariants (such as Betti numbers) in the limit of large sample sizes, it remains an open question whether the associated combinatorial complexes — such as the clique complex or Vietoris–Rips complex built on the CkNN graph — recover the full topological type of the manifold.

In this section, we take a step further and investigate whether CkNN-based simplicial complexes not only recover homology, but also become homotopy equivalent to the underlying manifold. We provide a partial result in this direction by proving that, under uniform sampling and appropriate parameter scaling, the clique complex built from a CkNN graph is homotopy equivalent to the manifold in the case of certain low-dimensional geometries.

To state our result precisely, we recall the notion of homotopy equivalence between topological spaces.

Definition 2.1 (Homotopy Equivalence). *Let X and Y be topological spaces. We say that X and Y are homotopy equivalent, written $X \simeq Y$, if there exist continuous maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that $g \circ f \simeq \text{id}_X$ and $f \circ g \simeq \text{id}_Y$, where $\simeq$ denotes homotopy of maps.*

2.1. Theorem: Homotopy Equivalence in Low Dimensions.

CONJECTURE 1 (Homotopy Equivalence for CkNN Complexes in Low Dimensions). *Let $\mathcal{M} \subset \mathbb{R}^D$ be a compact, connected, smooth, d -dimensional Riemannian manifold without boundary, and let $X_n \subset \mathcal{M}$ be an i.i.d. sample of n points from the uniform distribution on $\mathcal{M}$. Let $G_n = (X_n, E_n)$ be the CkNN graph constructed with fixed $k \in \mathbb{N}$ and $\delta > 0$, and let K_n denote the associated clique complex.*

Then there exists a value $\delta_0 > 0$ such that, for all $\delta \in (0, \delta_0)$, the complex $\mathcal{K}_n$ is homotopy equivalent to $\mathcal{M}$ with high probability as $n \rightarrow \infty$:

$$\mathbb{P}(\mathcal{K}_n \simeq \mathcal{M}) \rightarrow 1.$$

3. Worked Example: The Circle S^1

To illustrate our theorem, we consider the case where the underlying manifold is the unit circle $S^1 \subset \mathbb{R}^2$. Let $X_n \subset S^1$ be a set of n points sampled uniformly in angle from the circle. That is, each point has coordinates of the form

$$x_i = (\cos \theta_i, \sin \theta_i), \quad \theta_i \sim \text{Uniform}(0, 2\pi).$$

We construct the CkNN graph G_n on this point cloud using a fixed k and a scaling parameter $\delta > 0$. The edge condition

$$\|x_i - x_j\| < \delta \sqrt{r_k(x_i)r_k(x_j)},$$

adapts to local sampling density, though in this uniform case, $r_k(x_i) \approx r_k$ is nearly constant across all x_i .

As $n \rightarrow \infty$, the sample becomes dense on the circle, and for appropriately chosen δ , the resulting CkNN graph approximates a cycle. The clique complex $\text{Cl}(G_n)$ then becomes a triangulated polygon wrapping around the circle, and its homotopy type converges to that of S^1 .

This simple case demonstrates how the homotopy type of the manifold can be recovered from finite samples. Notably, a standard ε -ball graph with fixed radius may fail to capture the full loop if the point spacing is uneven, but the CkNN construction compensates automatically by scaling edge thresholds according to local density.

This example serves as a visual and intuitive foundation for the more general low-dimensional result in Theorem 1, and provides a natural testbed for simulations verifying the empirical convergence of homotopy type.

3.1. Relation to Existing Work. While persistent homology captures the multiscale evolution of homological features, it does not provide a homotopy-equivalent reconstruction of the manifold at any particular scale. Our result strengthens this connection by showing that a single, appropriately constructed CkNN complex can recover the full topological type in low-dimensional settings. This places CkNN closer in spirit to classical constructions like Čech and Vietoris–Rips complexes, which have known homotopy guarantees under metric-ball sampling, but extends these ideas to density-adaptive, graph-based methods.

4. Conclusion and Future Work

In this paper, we explored how the manifold hypothesis—a foundational assumption in geometric learning—can be rigorously realized through the Continuous k -Nearest Neighbors (CkNN) graph construction. We reviewed the theoretical foundations of CkNN as a consistent estimator of manifold geometry and topology, emphasizing its spectral convergence properties and its role in enabling topologically meaningful data analysis.

Building on this foundation, we proposed: under uniform sampling and appropriate scaling, the clique complex built from a CkNN graph recovers the full homotopy type of the underlying manifold in low-dimensional settings. This strengthens the

topological guarantees of CkNN beyond homology, establishing a deeper connection between discrete graph constructions and the intrinsic structure of sampled manifolds.

Future directions include extending this homotopy equivalence result to higher-dimensional and more general manifolds, incorporating non-uniform sampling densities, and exploring boundary-aware variants of CkNN. Additional work is also needed to connect these theoretical advances with practical algorithms for parameter selection, noise robustness, and large-scale implementation.

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